

Implementing a CI/CD Pipeline for GlobalMart E-Commerce Platform

Overview of Project

Scenario:

GlobalMart, a leading e-commerce retailer with 300+ employees across 3 regions, has implemented a CI/CD pipeline for their product catalog service, but is experiencing deployment failures and site outages. With an estimated \$25,000 in lost revenue per hour of downtime, they urgently need help troubleshooting their pipeline issues. As their Cloud Support Engineer, you'll identify and resolve the misconfiguration problems causing these failures.

Solution:

A fully automated CI/CD pipeline using AWS CodePipeline and CodeDeploy with comprehensive monitoring capabilities. This project demonstrates how AWS DevOps services work together and teaches essential troubleshooting skills for common CI/CD misconfigurations.

About Project:

I recreated GlobalMart's CI/CD pipeline with the same issues their team faced, then troubleshooted using CloudWatch, CodePipeline logs, and the AWS Console. I gained hands-on experience diagnosing and fixing real-world deployment issues — just as a cloud support engineer would in a production environment.

Steps Performed:

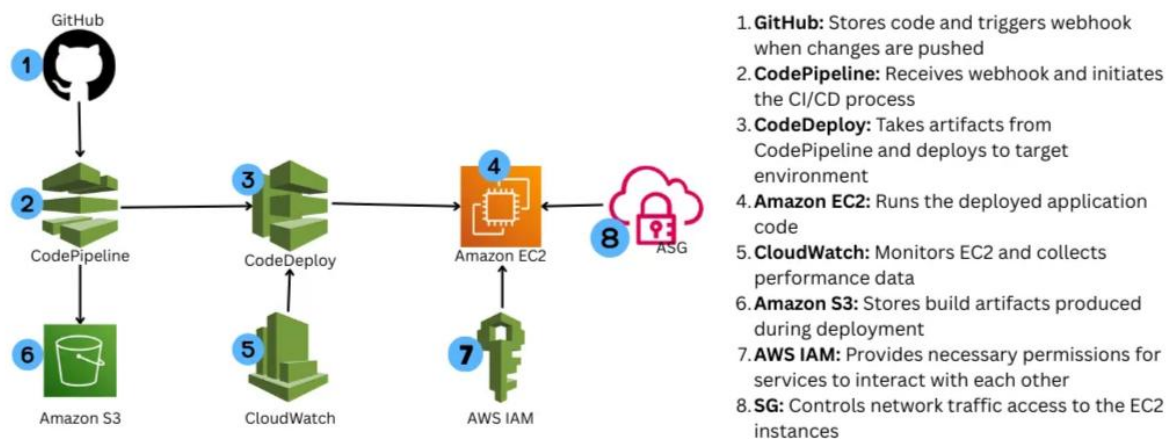
1. Setting Up Environment: Prepared EC2 and GitHub repository
2. Configuring AWS Services: Setting up IAM roles and deployment tools

3. Building CI/CD Pipeline: Configuring CodeDeploy and CodePipeline
4. Troubleshooting Pipeline Issues: Fixed common deployment problems

Services Used:

- AWS CodePipeline: Fully managed continuous delivery service
- AWS CodeDeploy: Deployment service for EC2 instances
- Amazon EC2: Compute instances for hosting the e-commerce application
- AWS IAM: Identity and access management for AWS resources
- GitHub: Source code repository with webhook integration
- Amazon S3: Storage for deployment artifacts
- AWS Security Groups: Virtual firewalls controlling network traffic

This is the architectural diagram for the project:



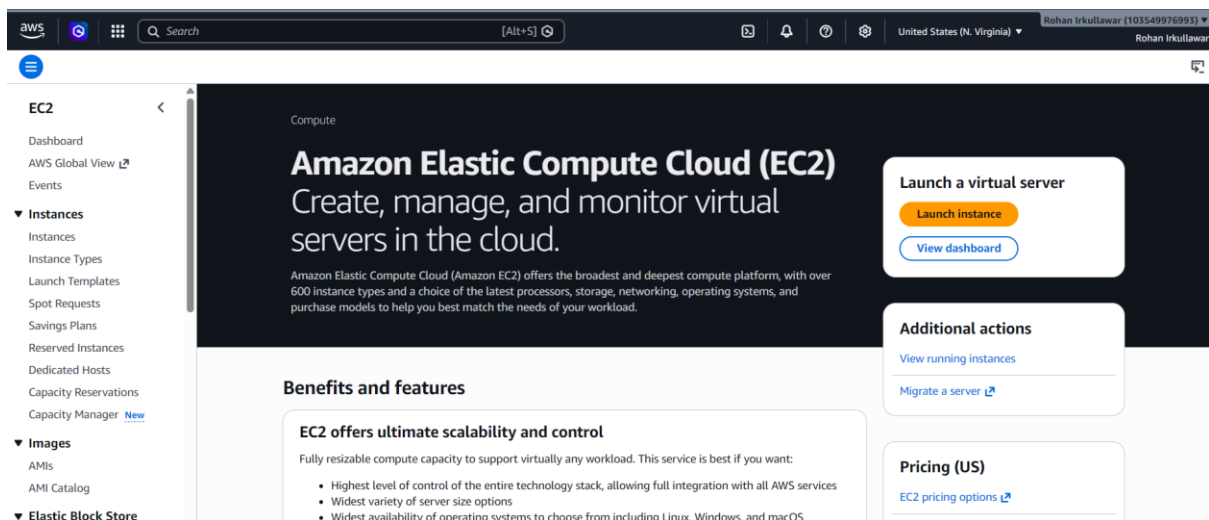
Infrastructure Setup: EC2 Configuration

Introduction:

In this hands-on project, I built, intentionally broke, and fixed a CI/CD pipeline for GlobalMart's e-commerce catalog. This first section walks through how I created a fully functional pipeline that deploys a React application to an EC2 instance.

Implementation Steps

Launched Instance:



Configured instance:

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

Globalmart-ec2

Add additional tags

▼ Name and tags

Key

Info

Q Name X

Value

Info

Q GlobalMart-WebServer X

Use: GlobalMart-WebServer

Resource types

Info

Select resource types

Instances X

Remove

Amazon Linux

aws

macOS

Mac

Ubuntu

Canonical Ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

>

🔍

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI

ami-00e801948462f718a (64-bit (x86), uefi-preferred) / ami-0b183bb1259186479 (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

▼ Instance type

Info | [Get advice](#)

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Linux base pricing: 0.0116 USD per Hour

On-Demand Windows base pricing: 0.0162 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour

On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand RHEL base pricing: 0.026 USD per Hour

🔍 All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login)

Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

kp-1

🔄 Create new key pair

Configured Network Settings:

- Used the default VPC and subnet
- Ensured "Auto-assign public IP" is set to "Enable"

▼ Network settings

Info

Network

Info

vpc-00ecc19fd940b1efc

Subnet

Info

No preference (Default subnet in any availability zone)

Auto-assign public IP

Info

Enable

Setting up Security Group:

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

- ☒ Allow SSH traffic from
Helps you connect to your instance
- ☒ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server
- ☒ Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

My IP
49.205.248.86/32

▼ Configure storage [Info](#)

1x GiB Root volume, Not encrypted

[Add new volume](#)

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

[↑ Choose file](#)

```
#!/bin/bash
exec > /var/log/user-data.log 2>&1
yum update -y

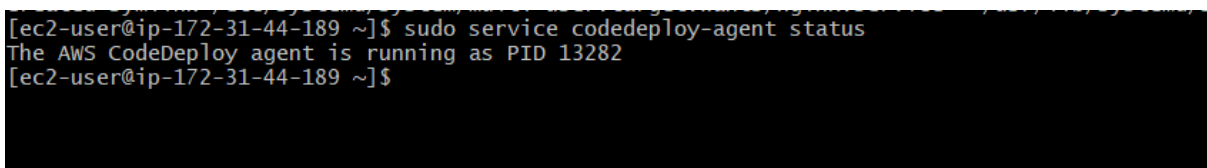
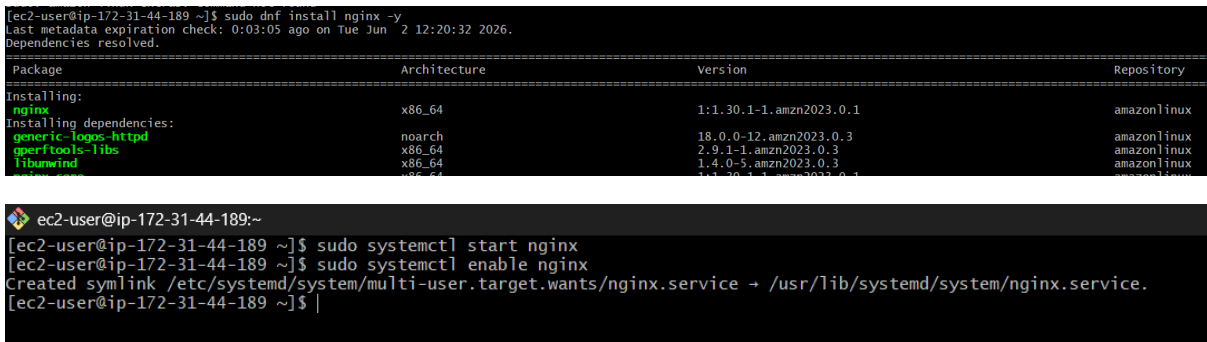
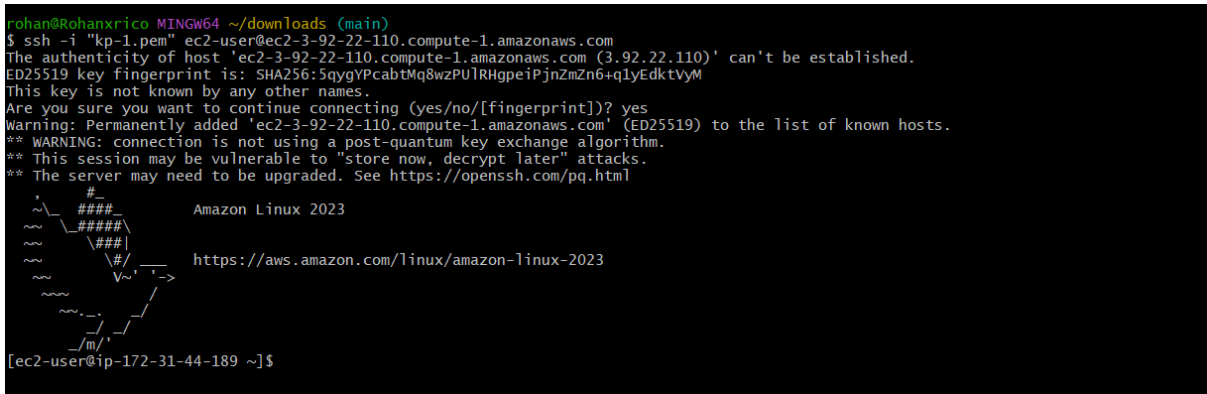
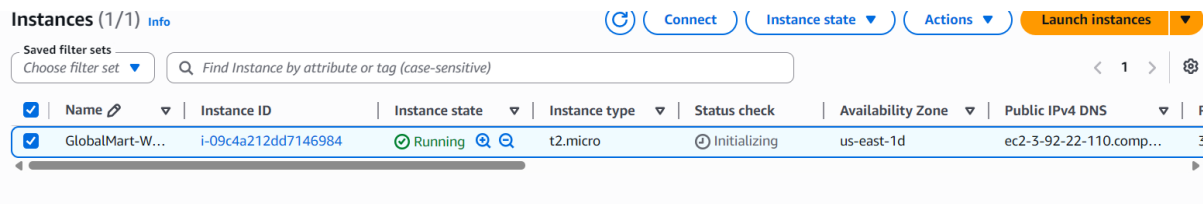
# Install Node.js v16
curl -sL https://rpm.nodesource.com/setup_16.x | bash -
yum install -y nodejs

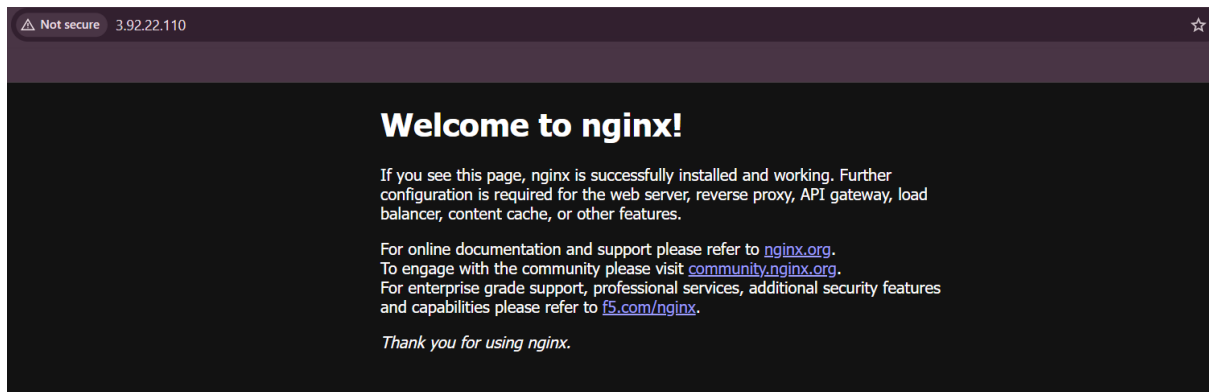
# Install CodeDeploy agent
yum install -y ruby wget
cd /home/ec2-user
wget https://aws-codedeploy-us-east-1.s3.amazonaws.com/latest/install
chmod +x ./install
./install auto
service codedeploy-agent start
```

This bootstrap script automatically runs when instance launches, installing required software without manual intervention. This is a key concept in Infrastructure as Code (IaC) and enables consistent, repeatable deployments.

The script installs: Node.js: The runtime environment needed to run our JS-based application

CodeDeploy agent: Required for AWS CodeDeploy to deploy applications to this instance





Verified Node Version:

```
ec2-user@ip-172-31-44-189:~  
[ec2-user@ip-172-31-44-189 ~]$ node --version  
v16.20.2  
[ec2-user@ip-172-31-44-189 ~]$ |
```

Source Control: GitHub Repository Configuration

Cloned repository to local machine:

Git clone <https://github.com/irkullawarrohan-hue/globalmart-catalog.git>

```
Windows PowerShell  
PS C:\Users\rohan\downloads\projects> git clone https://github.com/irkullawarrohan-hue/globalmart-catalog.git  
Cloning into 'globalmart-catalog'...  
remote: Enumerating objects: 11400, done.  
remote: Counting objects: 100% (401/401), done.  
remote: Compressing objects: 100% (262/262), done.  
Receiving objects: 94% (10716/11400), 48.07 MiB | 15.75 MiB/s
```

Build the React App Locally:

```
PS C:\Users\rohan\downloads\projects\globalmart-catalog\projects\cloud-support-engineer-projects\project-3\globalmart-ca  
talog> npm install  
npm warn deprecated stable@0.1.8: Modern JS already guarantees Array#sort() is a stable sort, so this library is depreca  
ted. See the compatibility table on MDN: https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Object  
s/Array/sort#browser_compatibility  
npm warn deprecated rollup-plugin-terser@7.0.2: This package has been deprecated and is no longer maintained. Please use  
@rollup/plugin-terser  
npm warn deprecated sourcemap-codec@1.4.8: Please use @jridgewell/sourcemap-codec instead
```

```
PS C:\Users\rohan\downloads\projects\globalmart-catalog\projects\cloud-support-engineer-projects\project-3\globalmart-ca
talog> npm run build
> ecommerce@0.1.0 build
> react-scripts build
```

Building locally allows you to verify the application works as expected before deployment. The build process optimizes the code, minifies files, and creates static assets that are ready for production serving, resulting in better performance for end users.

Modified .gitignore to Include build/ Folder:

```
# dependencies
node_modules
/.pnp
.pnp.js

# testing
/coverage

# production
#/build
█
# misc
.DS_Store
.env.local
.env.development.local
.env.test.local
.env.production.local

npm-debug.log*
yarn-debug.log*
yarn-error.log*
~
~
```


Then force added the build folder:

`git add -f build/`

```
PS C:\Users\rohan\downloads\projects\globalmart-catalog\projects\cloud-support-engineer-projects\project-3\globalmart-ca
talog> git add -f build
warning: in the working copy of 'projects/cloud-support-engineer-projects/project-3/globalmart-catalog/build/asset-manif
est.json', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'projects/cloud-support-engineer-projects/project-3/globalmart-catalog/build/static/css/
main.5e67c8af.css', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'projects/cloud-support-engineer-projects/project-3/globalmart-catalog/build/static/js/m
ain.e7776cb7.js', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'projects/cloud-support-engineer-projects/project-3/globalmart-catalog/build/static/js/m
ain.e7776cb7.js.LICENSE.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'projects/cloud-support-engineer-projects/project-3/globalmart-catalog/build/static/medi
a/fontawesome-webfont.c1e38fd9e0e74ba58f7a.svg', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\rohan\downloads\projects\globalmart-catalog\projects\cloud-support-engineer-projects\project-3\globalmart-ca
talog>
```

While best practices typically suggest excluding build artifacts from repositories, in this CI/CD implementation we're using a simplified approach where CodeDeploy directly deploys the pre-built files. In more sophisticated pipelines, the build would happen during the CI process instead.

Added deployment configuration files:

Created a file named `appspect.yml` in the root directory using

This file is required by AWS CodeDeploy to understand how to deploy your application. It defines which files should be copied where and which scripts to run at specific deployment lifecycle events

```

version: 0.0
os: linux
files:
  - source: /build
    destination: /usr/share/nginx/html/
hooks:
  BeforeInstall:
    - location: scripts/before_install.sh
      timeout: 300
      runas: root
  AfterInstall:
    - location: scripts/after_install.sh
      timeout: 300
      runas: root

```

The ***appspec.yml*** configuration instructs CodeDeploy to copy your build folder to Nginx's web directory and run specific scripts before and after installation. This ensures your web server is properly prepared before deployment and correctly configured afterward.

Created deployment scripts

Created scripts directory and two script files:

```

#!/bin/bash
# Ensure nginx is installed and deployment directory exists
if [ ! -d /usr/share/nginx/html ]; then
  mkdir -p /usr/share/nginx/html
fi
rm -rf /usr/share/nginx/html/*
~

```

Pasted in *scripts/before_install.sh*

The *before_install* script prepares the deployment environment by ensuring the destination directory exists and clearing any previous files.

```
#!/bin/bash
# Configure Nginx for React SPA
cat > /etc/nginx/conf.d/react-app.conf << 'EOL'
server {
    listen 80;
    server_name _;
    root /usr/share/nginx/html;
    index index.html;

    location / {
        try_files $uri $uri/ /index.html;
    }
}
EOL

# Remove default config if it exists
if [ -f /etc/nginx/conf.d/default.conf ]; then
    rm -f /etc/nginx/conf.d/default.conf
fi

# Set proper permissions and restart nginx
chmod -R 755 /usr/share/nginx/html
systemctl restart nginx
```

Pasted in *scripts/after_install.sh*

The after_install script configures Nginx to properly serve your React single-page application

Made scripts executable and commit changes:

```
chmod +x scripts/*.sh
```

```
git add appspec.yml scripts/
```

```
git commit -m "Add deployment configuration for CI/CD pipeline"
```

```
git push origin main
```

Security: IAM Role Configuration

☒ **AWS service**

Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ **AWS account**

Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ **Web identity**

Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ **SAML 2.0 federation**

Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ **Custom trust policy**

Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

EC2

Choose a use case for the specified service.

Use case

☒ **EC2**

Allows EC2 instances to call AWS services on your behalf.

Filter by Type

Q AmazonS3ReadOnlyAccess X All types 1 match < 1 > ⚙

☒	Policy name ↗	Type	Description
☒	 AmazonS3ReadOnlyAccess	AWS managed	Provides read only access to all buckets v...

Filter by Type

Q AmazonSSMManagedInstanceCore X All types 1 match < 1 > ⚙

☒	Policy name ↗	Type	Description
☒	 AmazonSSMManagedInstanceCore	AWS managed	The policy for Amazon EC2 Role to enabl...

Filter by Type

Q AmazonEC2RoleforAWSCodeDeploy X All types 2 matches < 1 > ⚙

☒	Policy name ↗	Type	Description
☒	 AmazonEC2RoleforAWSCodeDeploy	AWS managed	Provides EC2 access to S3 bucket to dow...

This combination allows your EC2 instance to:

- Access deployment artifacts from S3 buckets (S3ReadOnlyAccess)
- Be managed remotely via AWS Systems Manager (SSMManagedInstanceCore)
- Interact with CodeDeploy service to receive deployments (EC2RoleforAWSCodeDeploy)

Role details

Role name

Enter a meaningful name to identify this role.

GlobalMart-EC2-Role

Maximum 64 characters. Use alphanumeric and '+=,.,@-_' characters.

Description

Add tags - optional [Info](#)

Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.

Key

Q Project

X

Value - optional

Use "GlobalMart"

Q GlobalMart

X

Remove

Created CodeDeploy Role:

☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

CodeDeploy

Choose a use case for the specified service.

Use case

☒ **CodeDeploy**
Allows CodeDeploy to call AWS services such as Auto Scaling on your behalf.

Permissions policies (1) [Info](#)

The type of role that you selected requires the following policy.

Policy name i	Type
AWSCodeDeployRole	AWS managed

Role details

Role name

Enter a meaningful name to identify this role.

GlobalMart-CodeDeploy-Role

Maximum 64 characters. Use alphanumeric and '+=,.,@-_' characters.

Description

Key

Q project

X

Value - optional

Q GlobalMart

X

Remove

Attached EC2 Role to your instance

Modify IAM role [Info](#)

Attach an IAM role to your instance.

Instance ID

 i-09c4a212dd7146984 (GlobalMart-WebServer)

IAM role

Select an IAM role to attach to your instance or create a new role if you haven't created any. The role you select replaces any roles that are currently attached to your instance.

GlobalMart-EC2-Role

Deployment: AWS CodeDeploy Setup

Created CodeDeploy Application:

Create application

Application configuration

Application name

Enter an application name

GlobalMart-Catalog

100 character limit

Compute platform

Choose a compute platform

EC2/On-premises

Tags

Key

prproject

Value - optional

GlobalMart

[Remove tag](#)

Created Deployment Group

Deployment group name

Enter a deployment group name

GlobalMart-Production

100 character limit

Use entered value: arn:aws:iam::103549976993:role/GlobalMart-CodeDeploy-Role

GlobalMart-CodeDeploy-Role

[arn:aws:iam::103549976993:role/GlobalMart-CodeDeploy-Role](#)

[X](#)

Tag group 1

Key

[X](#)

Value - optional

[X](#)

[Remove tag](#)

Deployment type

Choose how to deploy your application

☒ In-place

Updates the instances in the deployment group with the latest application revisions. During a deployment, each instance will be briefly taken offline for its update

☐ Blue/green

Replaces the instances in the deployment group with new instances and deploys the latest application. After instances in the replacement environment are registered with a load balancer, instances from the environment are deregistered and can be terminated.

Environment configuration

Select any combination of Amazon EC2 Auto Scaling groups, Amazon EC2 instances, and on-premises instances to add to this deployment

☐ Amazon EC2 Auto Scaling groups

☒ Amazon EC2 instances

Deployment settings

Deployment configuration

Choose from a list of default and custom deployment configurations. A deployment configuration is a set of rules that determines how fast an application is deployed and the success or failure conditions for a deployment.

CodeDeployDefault.AllAtOnce



Pipeline: AWS CodePipeline Configuration

Created Pipeline

Choose creation option [Info](#)

Step 1 of 7

Category

☐ Deployment

☐ Continuous Integration

☐ Automation

☒ Build custom pipeline

[Cancel](#)

[Next](#)

Pipeline settings

Pipeline name

Enter the pipeline name. You cannot edit the pipeline name after it is created.

No more than 100 characters

Execution mode [Info](#)

Choose the execution mode for your pipeline. This determines how the pipeline is run.

- ☐ Superseded
- ☒ Queued
- ☐ Parallel

Service role

☒ New service role
Create a service role in your account

☐ Existing service role
Choose an existing service role from your account

Role name

Type your service role name

- ☒ Allow AWS CodePipeline to create a service role so it can be used with this new pipeline

The queued execution mode ensures that only one pipeline execution runs at a time, preventing conflicts when multiple code changes are pushed rapidly.

Add source stage [Info](#)

Step 3 of 7

Source

Source provider

This is where you stored your input artifacts for your pipeline. Choose the provider and then provide the connection details.

Grant AWS CodePipeline access to your GitHub repository. This allows AWS CodePipeline to upload commits from GitHub to your pipeline.

[Connect to GitHub](#)

✔ You have successfully configured the action with the provider.



Repository

Q irkullawarrohan-hue/globalmart-catalog



Branch

Q main



☒ Enable automatic retry on stage failure

Build - *optional*

Build provider

Choose the tool you want to use to run build commands and specify artifacts for your build action.

☒ Commands

☐ Other build providers

Commands

Specify the shell commands to run with your compute action in CodePipeline. You do not need to create any resources in CodeBuild. Note: Using compute time for this action will incur separate charges in AWS CodeBuild.

npm install
npm run build

Environment variables - *optional*

Key value pair that is supplied to actions with managed compute.

Add environment variable

VPC ID

Specify the VPC ID that your compute action will use.



► Additional configuration

Environment type, compute

Input artifacts

Choose an input artifact for this action. [Learn more](#)

Add deploy stage [Info](#)

Step 6 of 7

Deploy - *optional*

Deploy provider

Choose how you want to deploy your application or content. Choose the provider, and then provide the configuration details for that provider.

AWS CodeDeploy



SourceArtifact 

Defined by: Source

No more than 100 characters

Application name
Choose an application that you have already created in the AWS CodeDeploy console. Or create an application in the AWS CodeDeploy console and then return to this task.

Q GlobalMart-Catalog 

Deployment group
Choose a deployment group that you have already created in the AWS CodeDeploy console. Or create a deployment group in the AWS CodeDeploy console and then return to this task.

Q GlobalMart-Production 

☒ Configure automatic rollback on stage failure

Reviewed and created pipeline:

Review [Info](#)

Step 7 of 7

Step 2: Choose pipeline settings

Pipeline settings

Pipeline name

GlobalMart-Pipeline

Pipeline type

V2

Execution mode

QUEUED

Artifact location

A new Amazon S3 bucket will be created as the default artifact store for your pipeline

Step 3: Add source stage

Source action provider

Source action provider

GitHub (via OAuth app)

PollForSourceChanges

false

Repo

globalmart-catalog

Owner

Snigdha-Chaudhari

Branch

main

Enable automatic retry on stage failure

Step 4: Add build stage

Build action provider

Build action provider

Commands

Commands

```
[{"version": "0.2", "phases": [{"install": {"runtime-versions": {"nodejs": "16"}, "pre_build": {"commands": ["- npm install", "build"], "commands": ["- npm run build"], "artifacts": {"files": ["- build/**/*", "- appspec.yml", "- scripts/**/*"]}]}
```

Enable automatic retry on stage failure

Enabled

Step 6: Add deploy stage

Deploy action provider

Deploy action provider

AWS CodeDeploy

ApplicationName

GlobalMart-Catalog

DeploymentGroupName

GlobalMart-Production

Configure automatic rollback on stage failure

Enabled

Enable automatic retry on stage failure

Disabled

Cancel

Previous

Create pipeline

[IAM](#) > [Roles](#) > [AWSCodePipelineServiceRole-us-east-1-GlobalMart-Pipeline](#) > Add permissions

Attach policy to AWSCodePipelineServiceRole-us-east-1-GlobalMart-Pipeline

► Current permissions policies (3)

Other permissions policies (1052)

Filter by Type		1 match	
awscodedeployfull		All types	
☐ Policy name	☐ Type	Description	
☐ AWSCodeDeployFullAccess	AWS managed	Provides full access to CodeDeploy resou...	

[Developer Tools](#) > [CodePipeline](#) > [Pipelines](#) > GlobalMart-Pipeline

GlobalMart-Pipeline

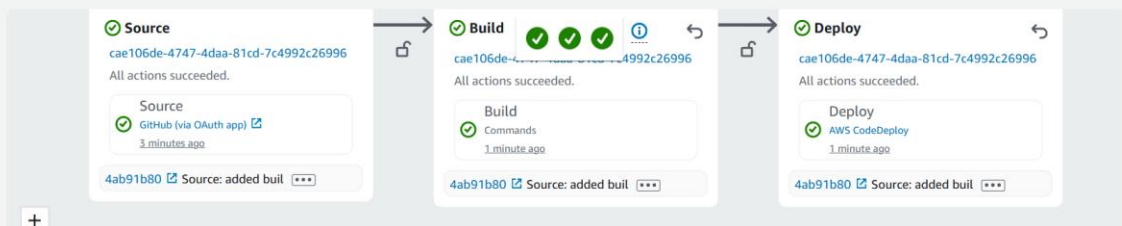
Edit

Stop execution

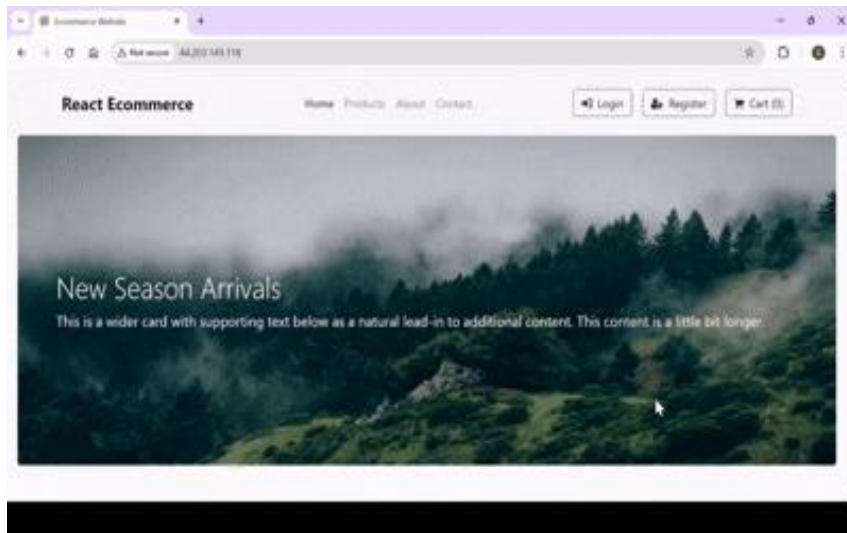
Clone pipeline

Release change

[Pipeline](#) | [Executions](#) | [Settings](#) | [Tags](#) | [Stage](#)



Verified deployment:



CI/CD Pipeline Troubleshooting

Introduction

Now that I built a working CI/CD pipeline, this section covers the common issues I encountered and worked through — just as a DevOps engineer would in real-world scenarios. By intentionally "breaking" components of the pipeline and then resolving them, I gained valuable troubleshooting experience.

Important Note: Before proceeding, I took notes about my working configuration and created screenshots of key settings. This helped me remember the correct configuration while troubleshooting.

Methodical Troubleshooting Approach When troubleshooting CI/CD pipeline issues, I followed these general steps:

1. Identified the failure point in the pipeline
2. Checked the logs for specific error messages
3. Verified configurations of the relevant AWS services
4. Tested my fix to ensure it resolved the issue
5. Documented the solution for future reference

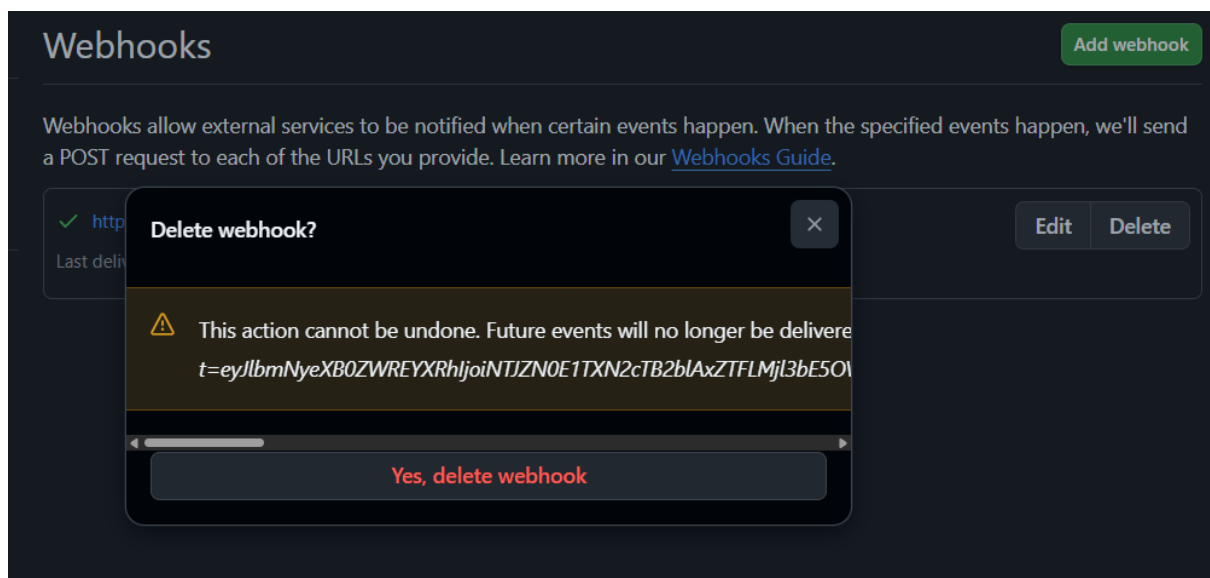
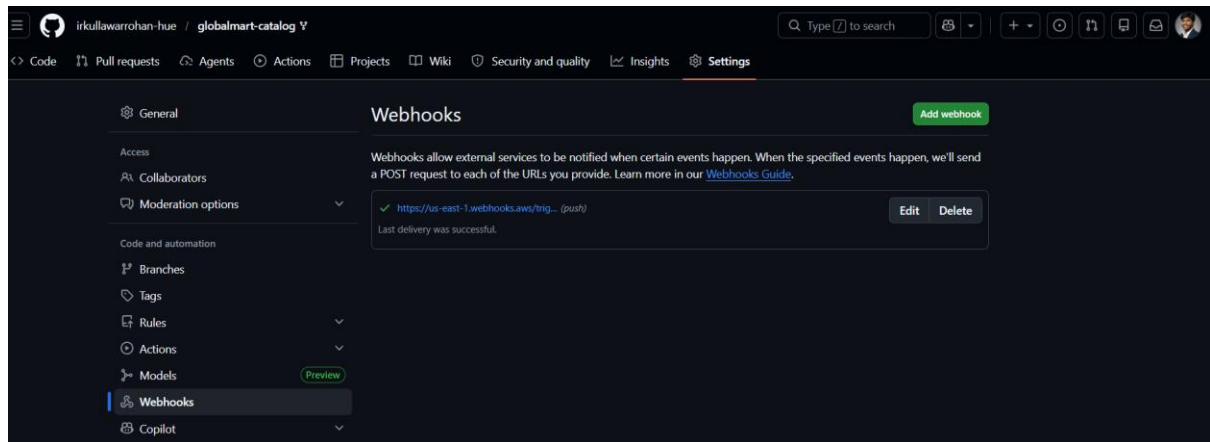
Issue 1: GitHub Webhook Connection

Breaking the Pipeline:

I navigated to my GitHub repository -

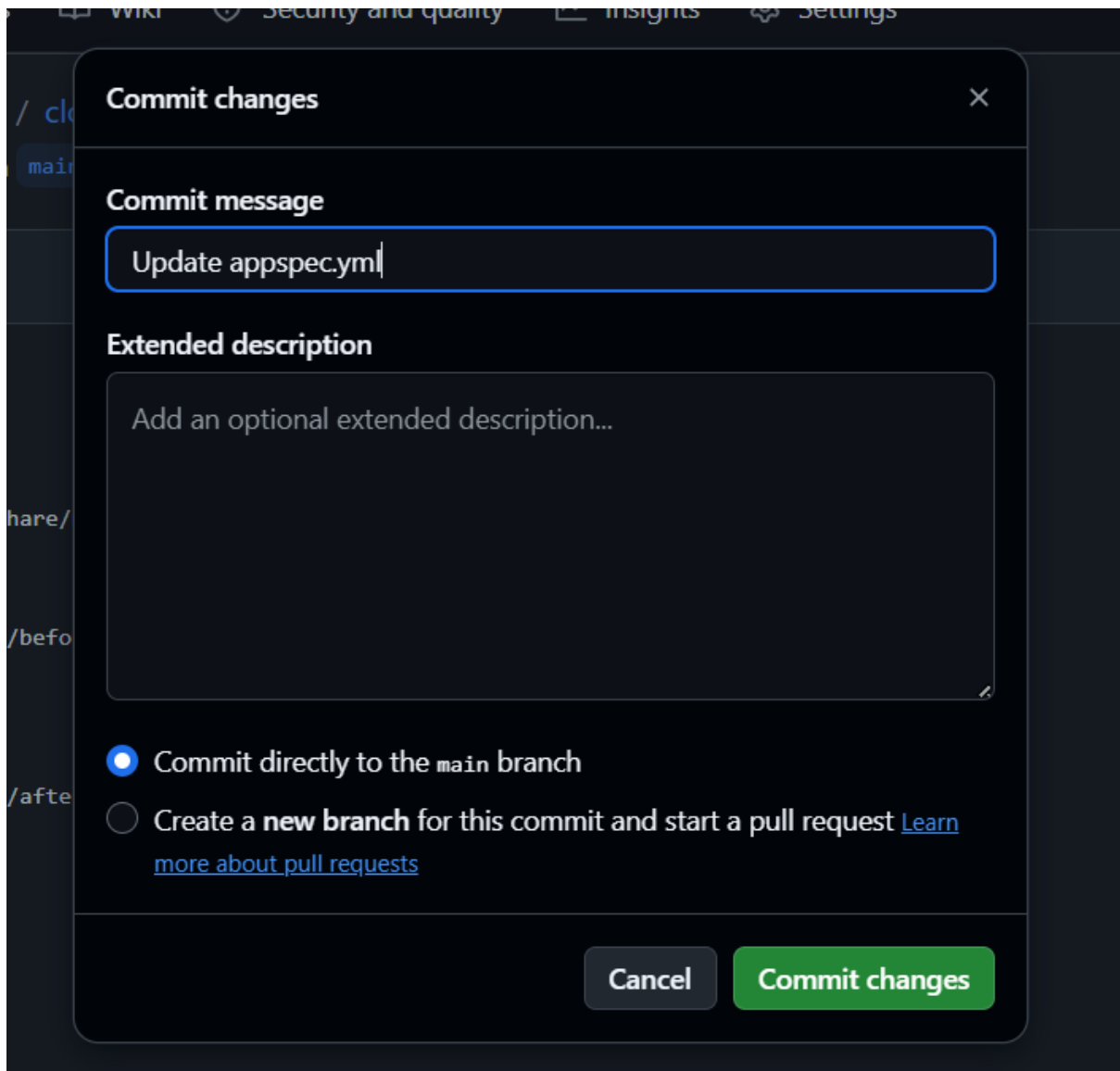
- Went to my forked repository on GitHub
- Clicked on "Settings" tab
- Selected "Webhooks" from the left sidebar
- Found the webhook created by AWS CodePipeline (it had an AWS URL)
- Clicked on the webhook, scrolled down and clicked "Delete"
- Confirmed deletion

By deleting this webhook, I broke the automatic communication link between my code repository and the deployment pipeline.



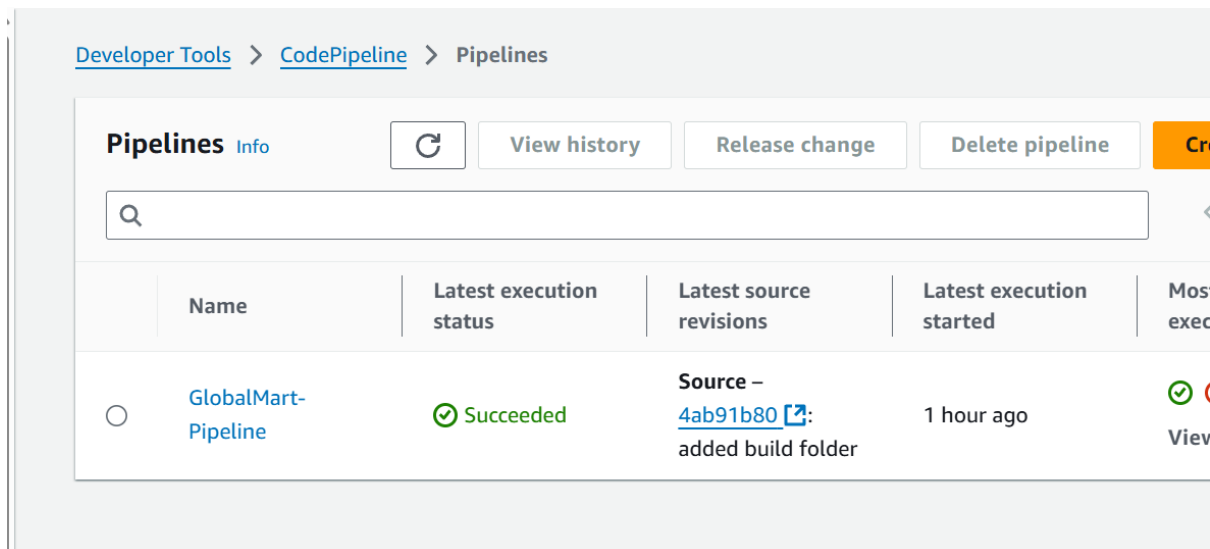
I tested the broken webhook:

- Made a small change to a file in my repository (changed version to 0.1 from 0.0 in appspec.yml)
- Committed and pushed the change
- Went to AWS CodePipeline and observed that the pipeline did not automatically start



Symptoms:

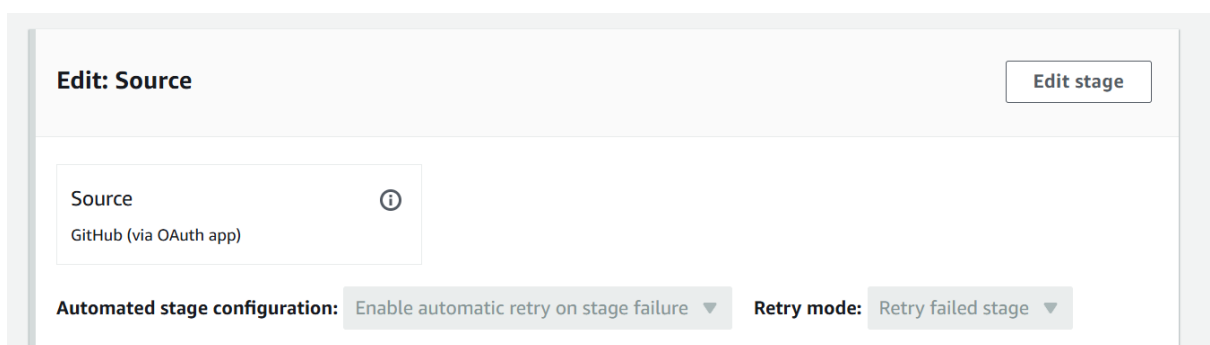
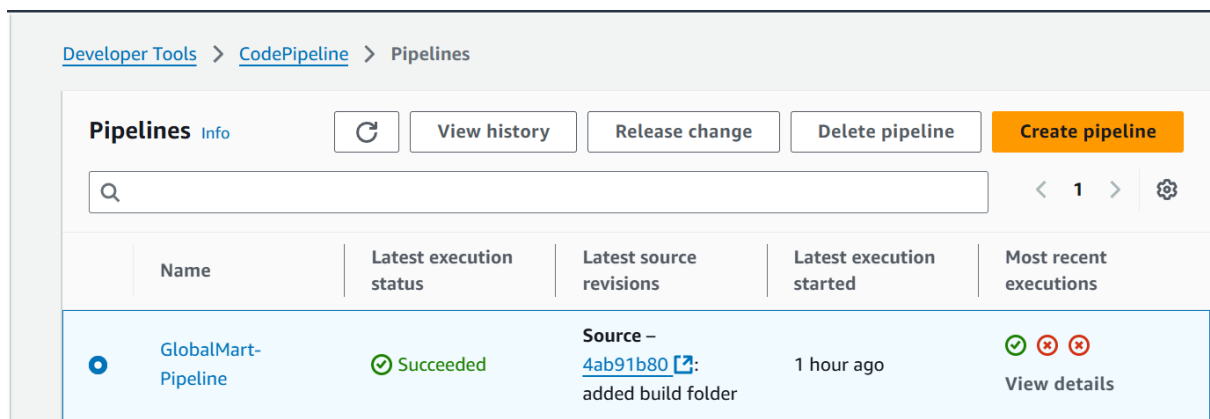
- Changes pushed to GitHub didn't trigger the pipeline
- No new executions appeared in CodePipeline history



Diagnosis:

I checked pipeline configuration:

- Went to CodePipeline > my pipeline
- Clicked "Edit" and reviewed the Source stage settings

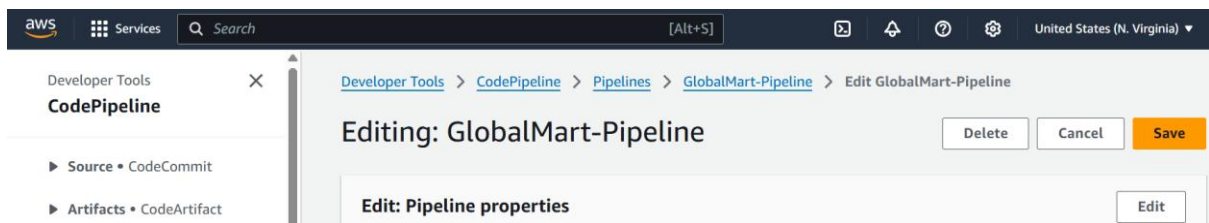


I checked GitHub repository settings:

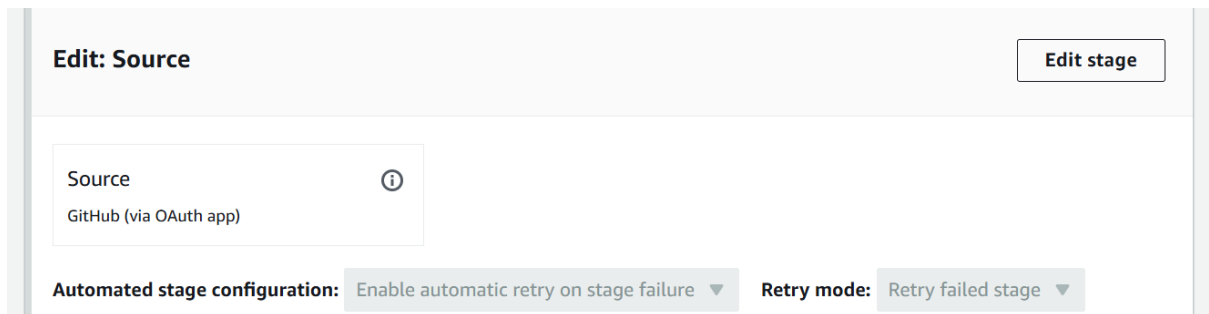
- Went to repository > Settings > Webhooks
- Confirmed no AWS webhook existed — this confirmed the webhook issue

Solution:

Went to CodePipeline > GlobalMart-Pipeline > Edit



Clicked "Edit" on the Source stage



Clicked "Add action"

Edit: Source

CancelDone

+ Add action group

Source ⓘ

GitHub (via OAuth app)

✎ ✕

+ Add action

+ Add action group

Automated stage configuration: Enable automatic retry on stage failure ▼Retry mode: Retry failed stage ▼

Provided Action name: webhook-creation2

Edit action

Action name

Choose a name for your action

webhook-creation2

No more than 100 characters

Selected Action provider: "GitHub (via OAuth app)"

Clicked "Connect to GitHub" > "Confirm"

No more than 100 characters

Action provider

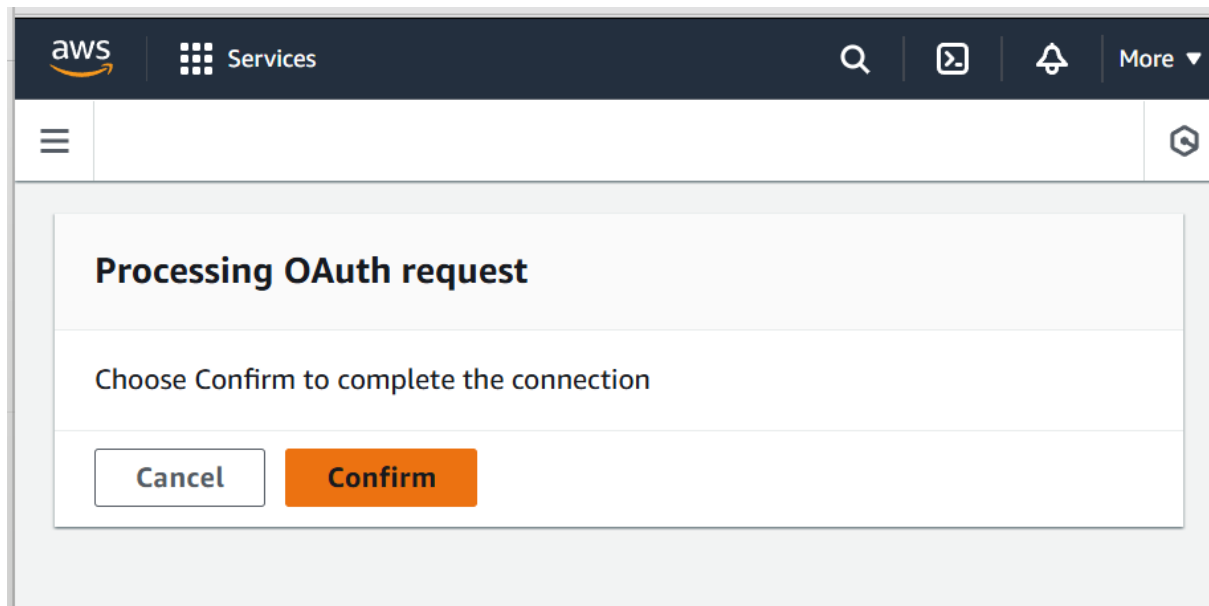
GitHub (via OAuth app) ▼

Grant AWS CodePipeline access to your GitHub repository. This allows AWS CodePipeline to upload commits from GitHub to your pipeline.

Connect to GitHub

ⓘ

The GitHub (via OAuth app) action is not recommended
The selected action uses OAuth apps to access your GitHub repository. This is no longer the recommended method. Instead, choose the GitHub (via GitHub App) action to access your repository by creating a connection. Connections use GitHub Apps to manage authentication and can be shared with other resources. [Learn more](#)



Selected my repository and branch:

Repository

Q irkullawarrohan-hue/globalmart_catalog X

Branch

Q main X

Provided artifact name: webhook-c2

Output artifacts

Choose a name for the output of this action.

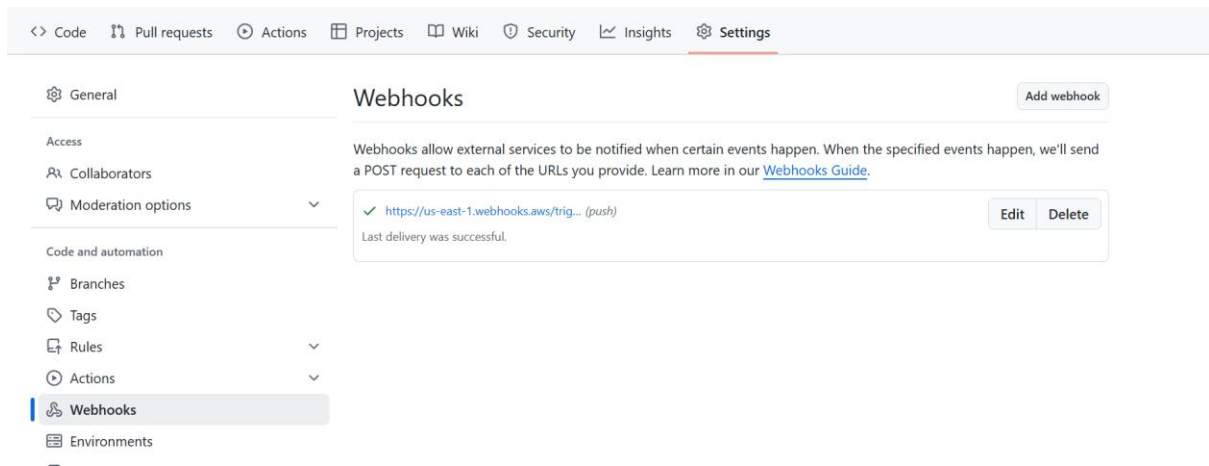
webhook-c2

No more than 100 characters

Clicked "Done" then "Save" — this recreated the webhook automatically

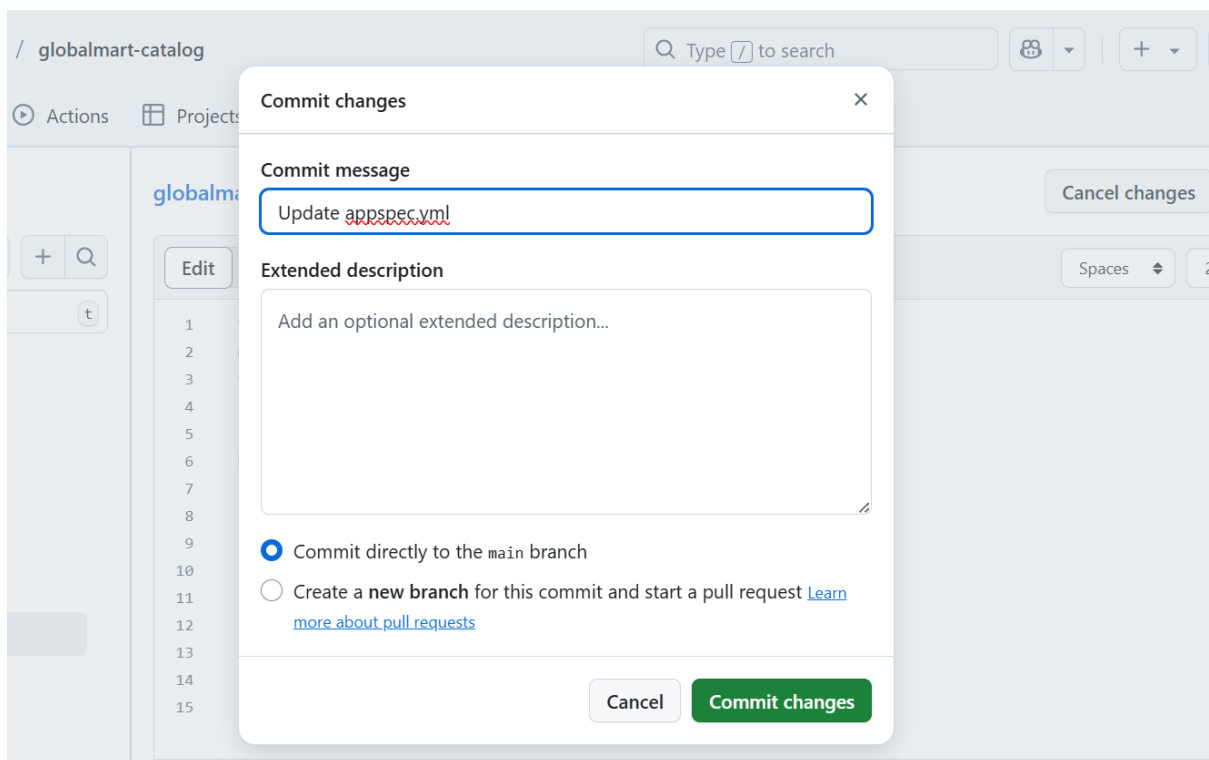
I verified webhook creation:

- Went to GitHub repository > Settings > Webhooks
- Confirmed a new webhook from AWS was created

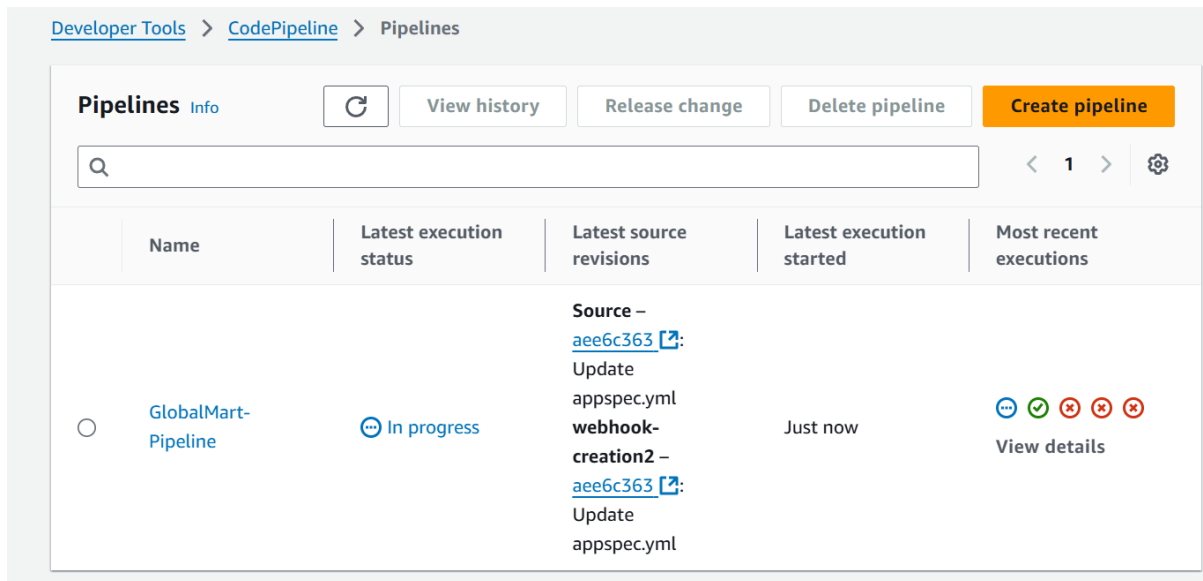


I tested the fix:

- Made a small change, committed and pushed
- Verified the pipeline triggered automatically



Verified the pipeline triggers automatically:

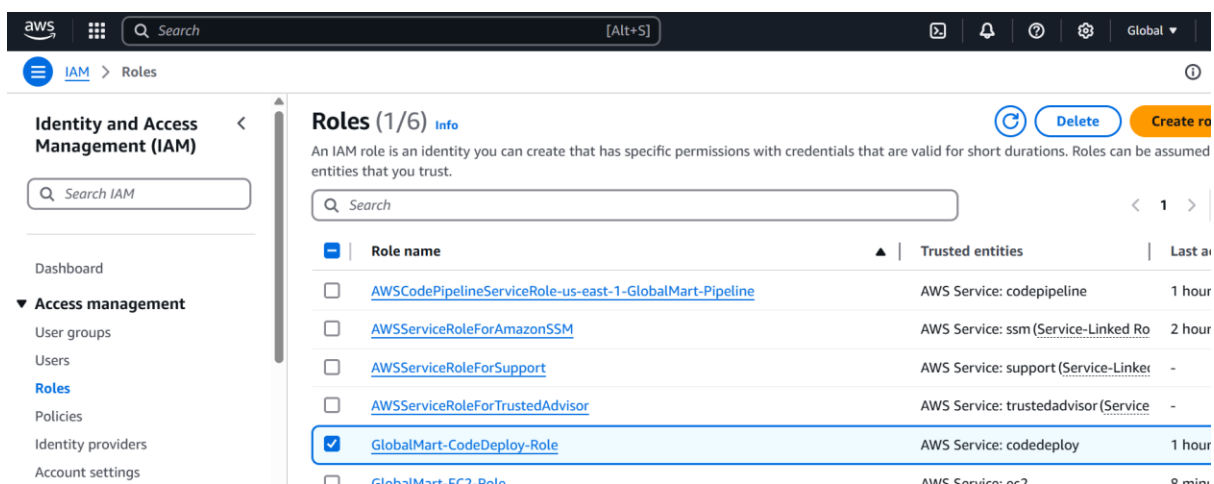


Issue 2: IAM Permissions Problem

I navigated to IAM in the AWS Console:

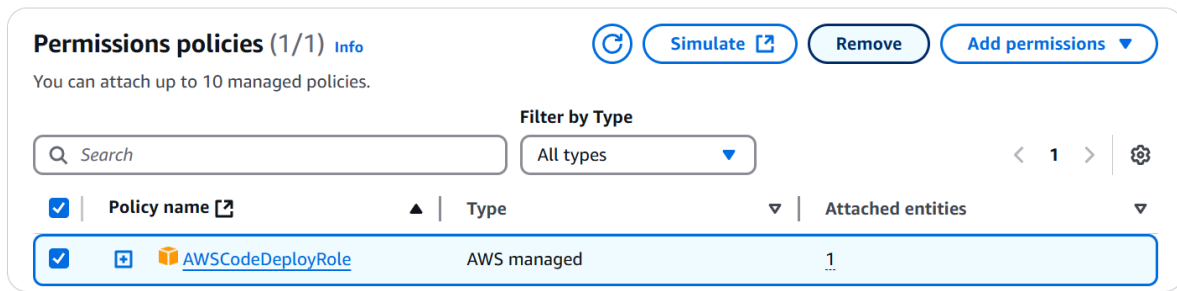
-Went to IAM > Roles

-Searched for and selected GlobalMart-CodeDeploy-Role

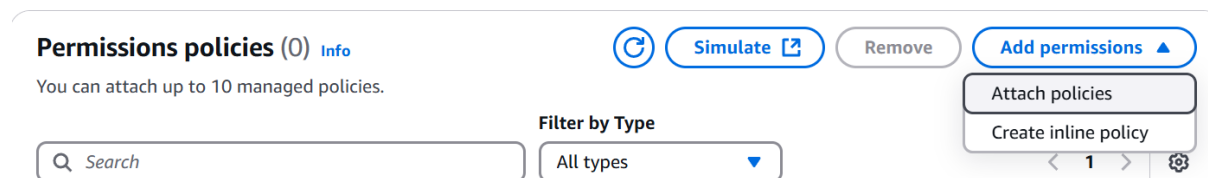


- Went to the "Permissions" tab

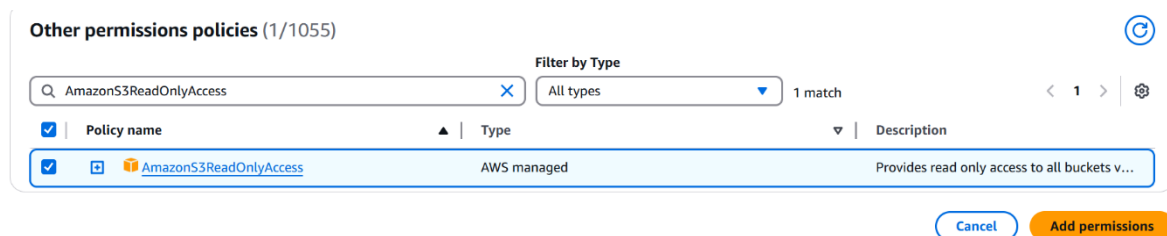
- Removed the AWSCodeDeployRole policy



Clicked "Add permissions" > "Attach policies"

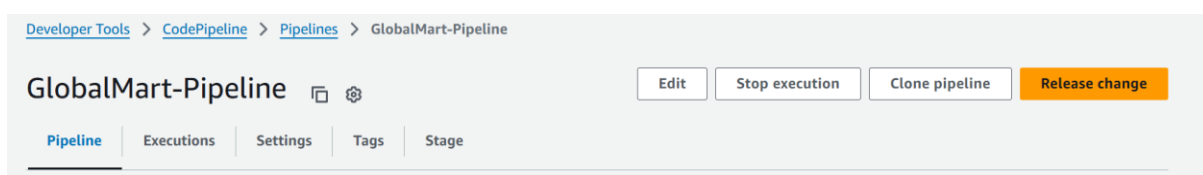


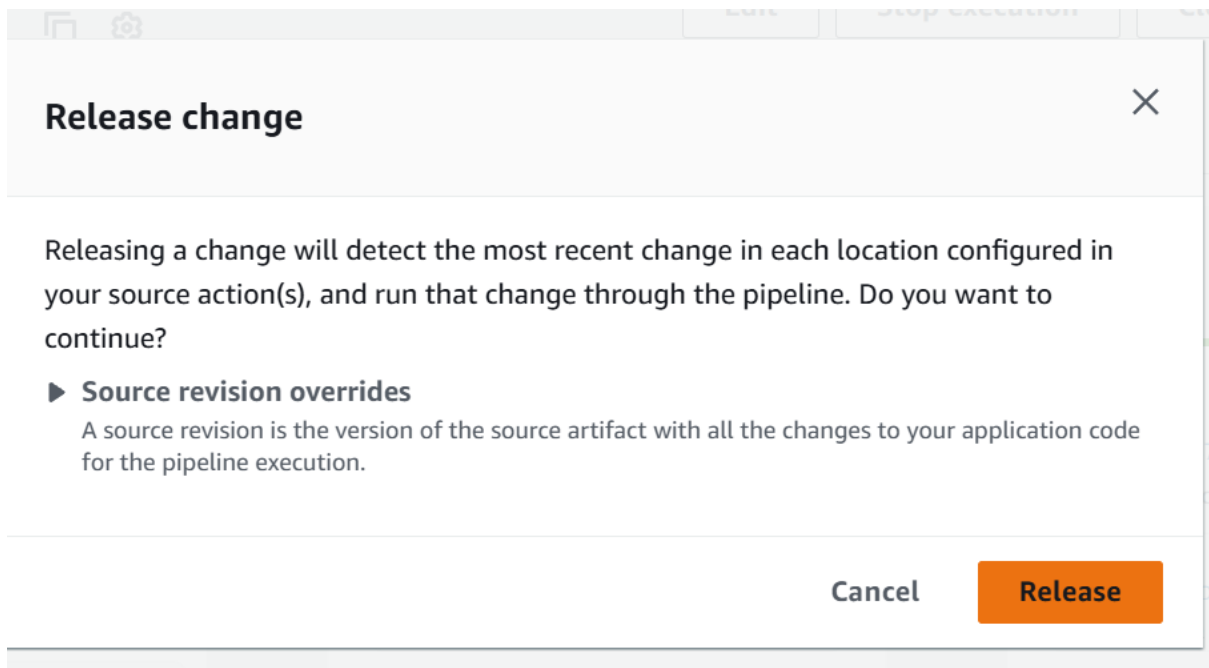
Searched for and attached AmazonS3ReadOnlyAc



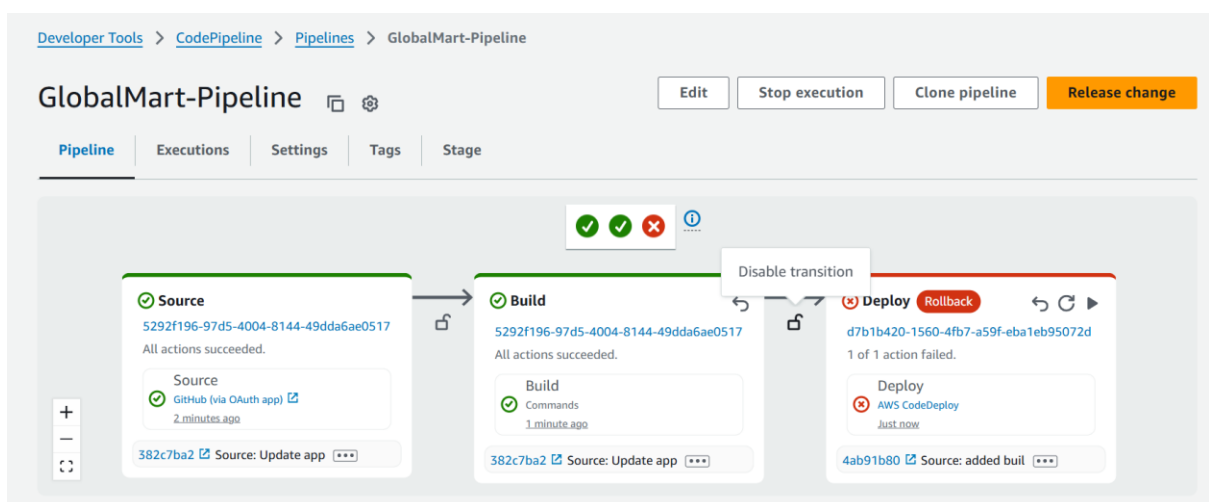
This severely restricted CodeDeploy to only reading from S3, preventing it from performing most of its core functions.

I tested the permission issue:





Observed that the pipeline failed during the Deploy stage



Reviewed the error messages in CodeDeploy

When a pipeline fails specifically at the Deploy stage (after successfully completing Source and Build stages), it suggests that the issue lies with the deployment service rather than with the code itself. Permission-related error messages in CodeDeploy logs further narrow down the problem to IAM configuration rather than application or network issues.

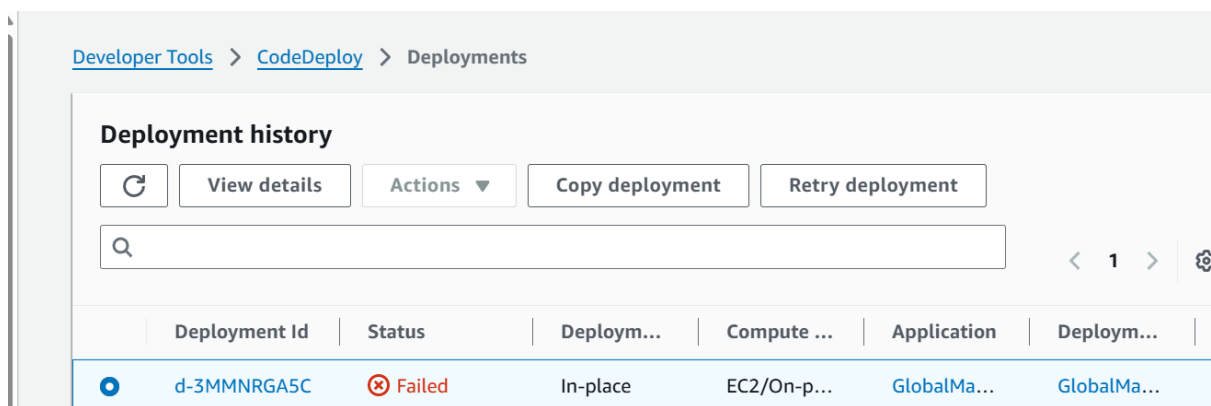
Symptoms:

- Pipeline fails at the Deploy stage
- CodeDeploy shows permission-related error messages

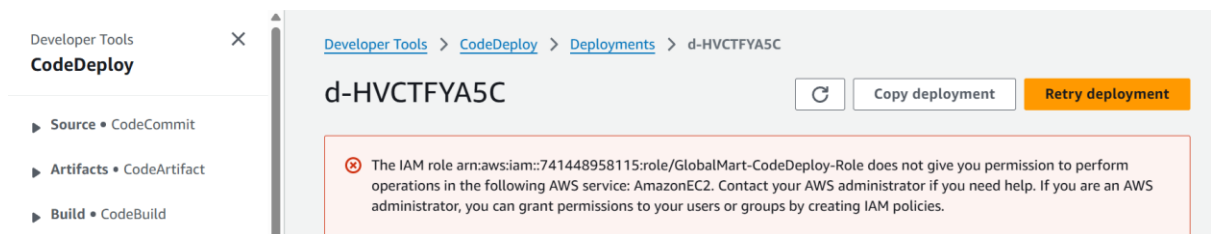
Diagnosis:

I checked CodeDeploy deployment logs:

- Went to CodeDeploy > Deployments
- Selected the failed deployment



- Reviewed error details — confirmed permission issues



I verified IAM role permissions:

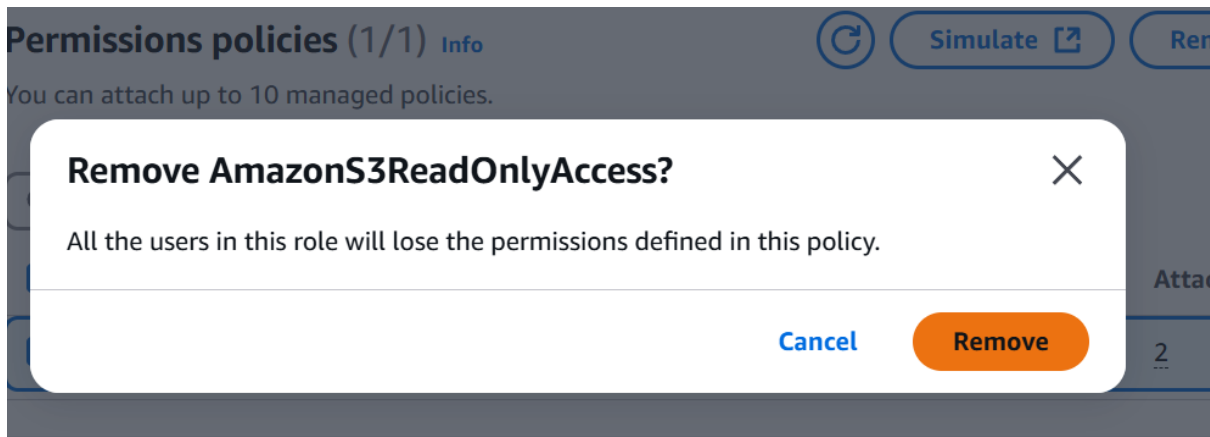
- Went to IAM > Roles > GlobalMart-CodeDeploy-Role
- Confirmed the proper policies were missing

Solution:

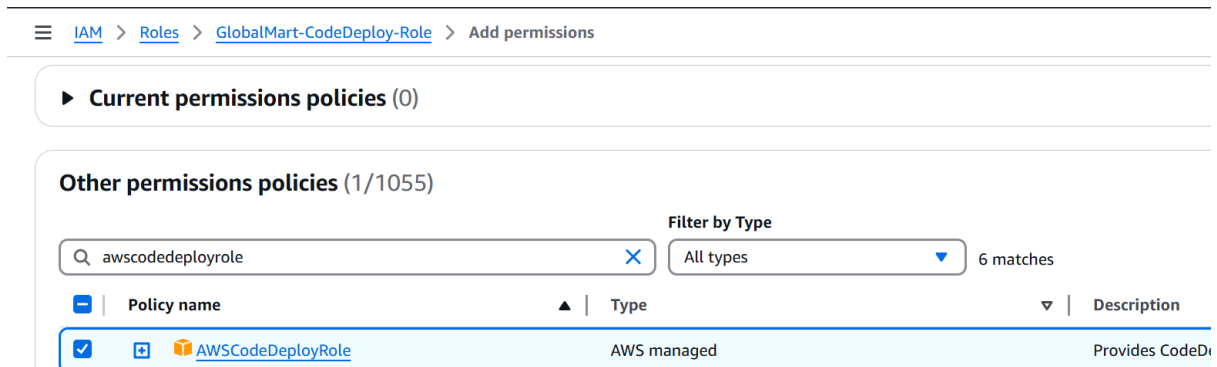
I updated the IAM role:

- Went to IAM > Roles > GlobalMart-CodeDeploy-Role

- Removed the restrictive AmazonS3ReadOnlyAccess policy

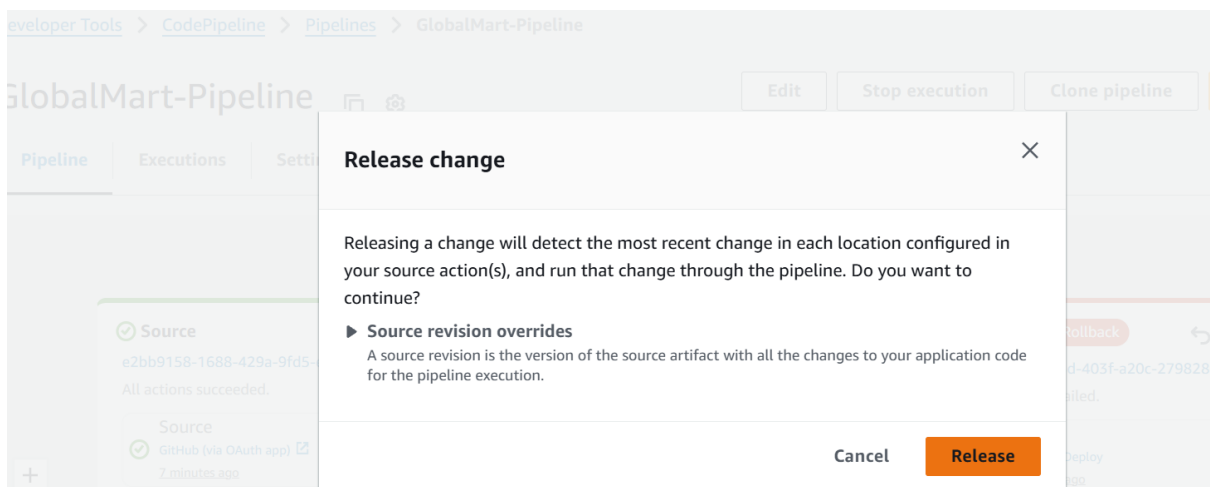


- Clicked "Add permissions" > "Attach policies"
- Searched for and attached the AWSCodeDeployRole policy

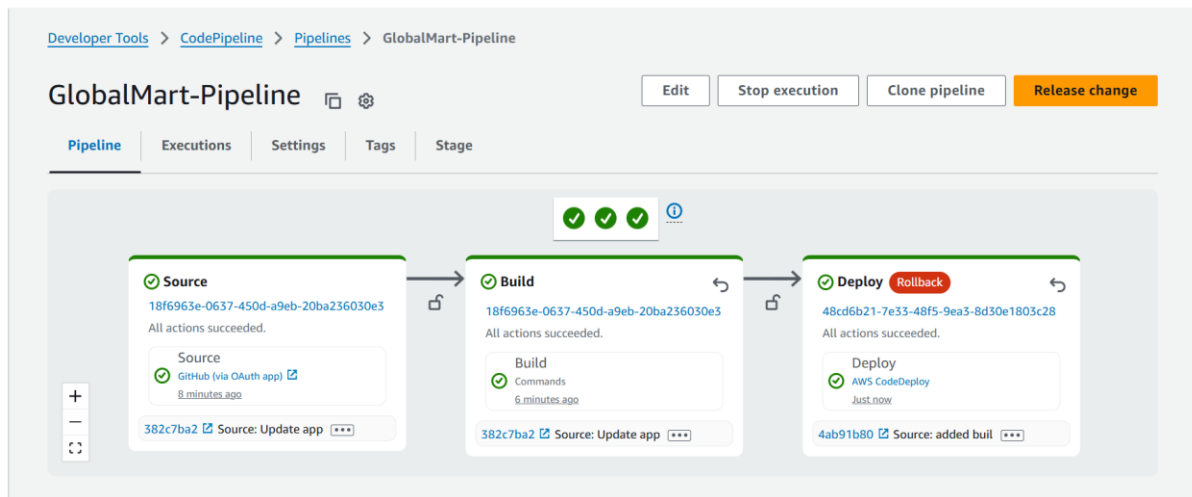


I tested the fix:

- In CodePipeline, clicked "Release change"



- Monitored the deployment progress — permission errors were resolved



Issue 3: Security Group Configuration

Breaking the Pipeline

I navigated to EC2 in the AWS Console:

- Went to EC2 > Security Groups
- Found the security group attached to my EC2 instance
- Selected the security group > "Inbound rules" tab

i-0c65e7cb847d84556 (GlobalMart-WebServer)

Security groups



[sg-06edee1da48f79404 \(launch-wizard-1\)](#)

▼ Inbound rules

- Clicked "Edit inbound rules"

Inbound rules	Outbound rules	Sharing - new	VPC associations - new	Tags
---------------	----------------	---------------	------------------------	------

Inbound rules (3)

Manage tags

Edit inbound rules

< 1 > ⚙

☐	Name	Security group rule ID	IP version	Type	Protocol
☐	-	sgr-07419eb6ae92e03ae	IPv4	HTTP	TCP
☐	-	sgr-02f9624cf9ad3a372	IPv4	SSH	TCP
☐	-	sgr-0a98122b067b26261	IPv4	HTTPS	TCP

- Deleted the rule allowing HTTP (port 80) traffic

Inbound rules

Info

Security group rule ID	Type	Protocol	Port range	Source	Description - optional	
sgr-02f9624cf9ad3a372	SSH	TCP	22	Cu...		Delete
sgr-0a98122b067b26261	HTTPS	TCP	443	Cu...		Delete
-	HTTP	TCP	80	An...		Delete

Add rule

- Clicked "Save rules"

I tested the security group issue:

- Tried to access the application at http://your-ec2-public-ip
- The connection timed out — confirmed the issue
- Noted that SSH access still worked, allowing remote troubleshooting

Symptoms:

- EC2 instance was running but not accessible via HTTP
- Browser connection timed out when trying to access the site

Diagnosis:

I checked EC2 instance status:

- Verified instance was running in the EC2 dashboard
- Tested SSH connection — confirmed basic connectivity

I checked security groups:

- Selected my EC2 instance > "Security" tab
- Reviewed inbound rules — confirmed HTTP rule was missing

Solution:

I updated the security group:

- Went to EC2 > Security Groups
- Selected the security group attached to my instance
- Clicked "Edit inbound rules"
- Added a new rule:
 - Type: HTTP
 - Protocol: TCP
 - Port range: 80
 - Source: 0.0.0.0/0
 - Description: "Allow HTTP traffic"
- Clicked "Save rules"

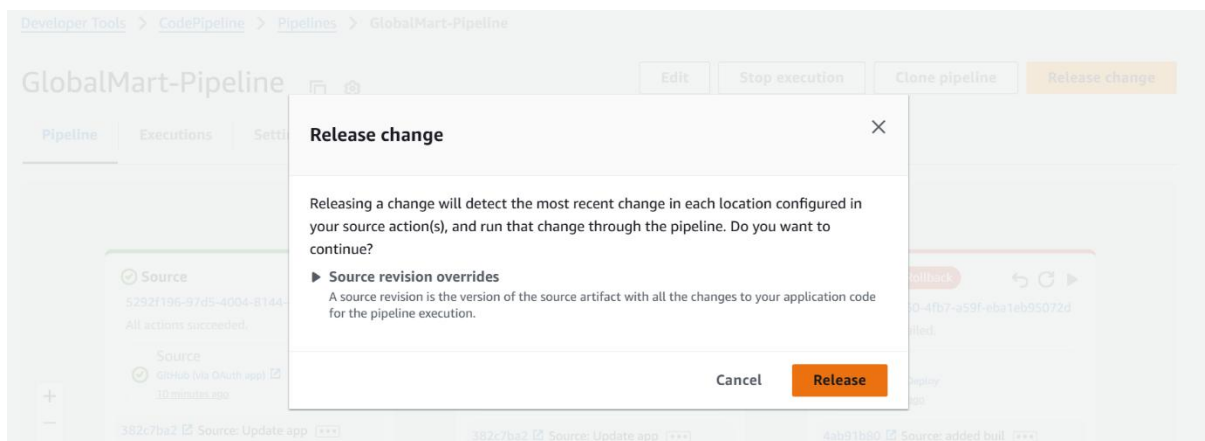
I tested the fix:

- Opened the browser and went to <http://your-ec2-public-ip>
- The website was now accessible

Issue 4: CodeDeploy Agent

Breaking the Pipeline

1. I connected to my EC2 instance via SSH:
 - `ssh -i your-key.pem ec2-user@my-public-ip`
2. I stopped the CodeDeploy agent:
 - `sudo service codedeploy-agent stop`
3. I tested the agent issue:
 - o In CodePipeline, clicked "Release change"



- o The deployment failed and timed out
- o In CodeDeploy, confirmed the agent was not responding

Symptoms:

- Deployment failed with agent communication errors
- CodeDeploy showed timeouts when trying to reach the instance

Diagnosis:

1. I checked agent status on EC2:
 - `ssh -i your-key.pem ec2-user@my-public-ip`
 - `sudo service codedeploy-agent status`

```
Stopping codedeploy-agent: ERROR: No AWS CodeDeploy agent running
```

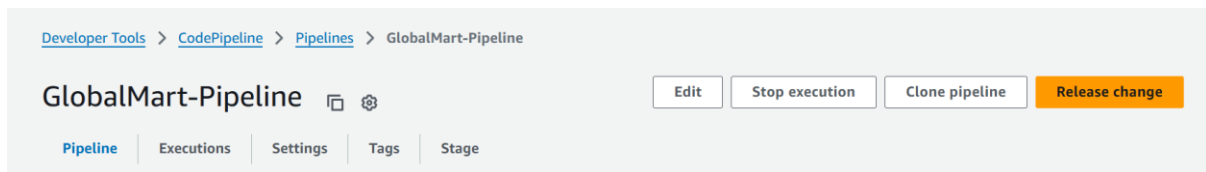
- Confirmed the agent was stopped

Solution:

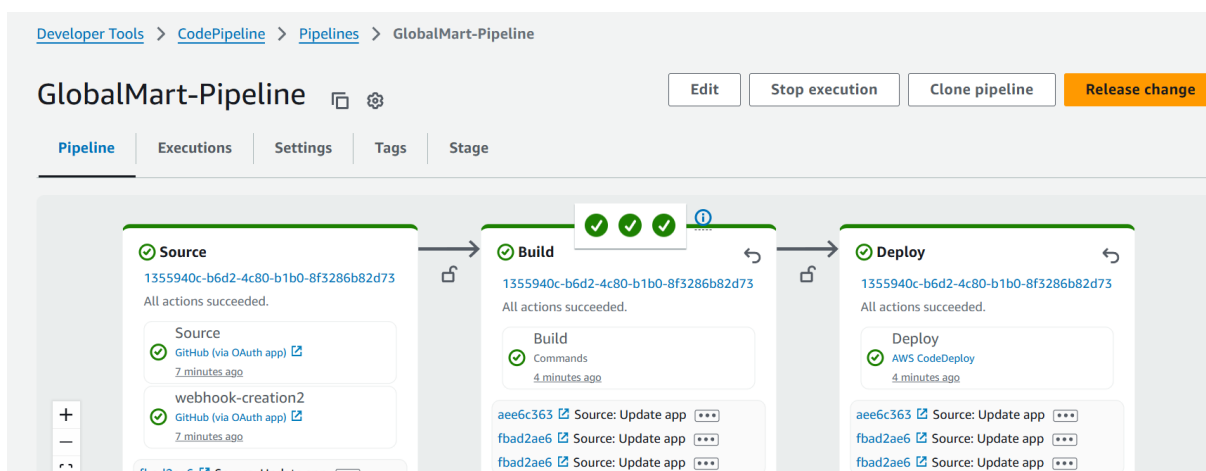
1. I started the CodeDeploy agent:
 - `sudo service codedeploy-agent start`
2. I verified the agent was running:
 - `sudo service codedeploy-agent status`
3. I configured auto-restart on reboot:
 - `sudo chkconfig codedeploy-agent on`

This ensured the CodeDeploy agent automatically starts whenever the instance reboots, adding resilience to the deployment pipeline.

4. I tested the fix:
 - In CodePipeline, clicked "Release change"



- The deployment communicated properly with the agent and deployed successfully
-



aws

Services

Search

[Alt+S]

United States (N. Virginia)

Developer Tools

CodeDeploy

Source • CodeCommit

Artifacts • CodeArtifact

Build • CodeBuild

Deploy • CodeDeploy

Getting started

Deployments

Deployment

Applications

Deployment configurations

On-premises instances

Pipeline • CodePipeline

Revision details

Revision location

Revision created

Revision description

s3://codepipeline-us-east-1-3cccf8607343-4f1e-b3fe-681586750b5d/GlobalMart-Pipeline/SourceArti/q2NhmnQ.zip?eTag=0526d7ebd87ee57c693f7663f5d36b5d-1

8 minutes ago

Application revision registered by
Deployment ID: d-56791LB5C

Deployment lifecycle events

< 1 >

Instance ID	Duration	Status	Most recent event	Events	Start time
i-0c65e7cb847d84556	6 seconds	Succeeded	ValidateService	View events	Apr 9, 2025 8:16 PM (UTC)